

Question 15.2

In the videos, we saw the “diet problem”. (The diet problem is one of the first large-scale optimization problems to be studied in practice. Back in the 1930’s and 40’s, the Army wanted to meet the nutritional requirements of its soldiers while minimizing the cost.) In this homework you get to solve a diet problem with real data. The data is given in the file `diet.xls`.

1. Formulate an optimization model (a linear program) to find the cheapest diet that satisfies the maximum and minimum daily nutrition constraints, and solve it using PuLP. Turn in your code and the solution. (The optimal solution should be a diet of air-popped popcorn, poached eggs, oranges, raw iceberg lettuce, raw celery, and frozen broccoli. UGH!)

If you want to see what a more full-sized problem would look like, try solving your models for the file `diet_large.xls`, which is a low-cholesterol diet model (rather than minimizing cost, the goal is to minimize cholesterol intake). I don’t know anyone who’d want to eat this diet – the optimal solution includes dried chrysanthemum garland, raw beluga whale flipper, freeze-dried parsley, etc. – which shows why it’s necessary to add additional constraints beyond the basic ones we saw in the video!

[Note: there are many optimal solutions, all with zero cholesterol, so you might get a different one. It probably won’t be much more appetizing than mine.]

Methodology

- I. First, I loaded the dataset. The dataset includes dollar signs, so I removed those dollar signs by using `.replace` and converted the cost to numeric. Then, I renamed the columns.
- II. Next, I used `.iloc` to split off the first 64 rows as actual food items. I used the last rows to get minimum and maximum nutrient requirements and assigned them to variables `min_row` and `max_row`. I did this to set constraints.
- III. I created dictionaries for the foods and nutrients. I extracted the list of food names from the dataframe and used `zip()` to pair each food name with its cost and turn those pairs into a dictionary. Then, I got the names of all nutrient columns and built a dictionary where the keys are the food names and the values are the amount of that nutrient in 1 serving. After that, I extracted the minimum and maximum required amounts for each nutrient.
- IV. Then, I defined the decision variables by using `Lpvariable.dicts`. `x` is the number of servings of each food and `y` is the binary variable of whether food is selected or not. We do this to minimize the cost using `x` and to set rules using `y`.
- V. To minimize the cost, I added up all the servings multiplied by their cost, which is the total cost of the diet. I used `lpSum` to do this.
- VI. Finally, I used a `lpSum()` inside a for loop to see when it is between the minimum and maximum requirement of nutrient. I did this to make the diet nutritionally balanced.

Results

Optimal solution found!

Frozen Broccoli: 0.26 servings
Celery, Raw: 52.64 servings
Lettuce, Iceberg, Raw: 63.99 servings
Oranges: 2.29 servings
Poached Eggs: 0.14 servings
Popcorn, Air-Popped: 13.87 servings

Total Cost: \$4.34

As a result, the optimal solution is when the value is \$4.34. This is the minimum total cost to meet all nutrient constraints. The cheapest combinations of foods that satisfies all daily nutrient minimum and maximum requirements is:

1. Frozen Broccoli: 0.26 servings
2. Celery, Raw: 52.64 servings
3. Lettuce, Iceberg, Raw: 63.99 servings
4. Oranges: 2.29 servings
5. Poached Eggs: 0.14 servings
6. Popcorn, Air-popped: 13.87 servings

Discussion of Results

According to the results, we need 52.64 servings of celery and 63.99 servings of lettuce, which is excessive but necessary because they are cheap and provide fiber, vitamins, and bulk with not many calories or fat. We need only 0.14 servings of poached eggs that hits the protein minimum and adds cholesterol and fat. This is minimized because of the higher cost of eggs. We need 13.87 servings of popcorn, which provides more calories, carbs, and fiber. This helps hit the carbohydrate minimum. We need 2.29 servings of oranges to provide the vitamin C, sugar, carbohydrates, calcium, and fiber. Finally, we need 0.26 servings of frozen broccoli to provide the protein, calcium, and vitamin C. This model minimized the cost while meeting the nutrient targets; however, it gave a diet that may be difficult to follow. Therefore, we will add more constraints in part 2 to make a more realistic and balanced diet.

2. Please add to your model the following constraints (which might require adding more variables) and solve the new model:
 - a. If a food is selected, then a minimum of 1/10 serving must be chosen. (Hint: now you will need two variables for each food i : whether it is chosen, and how much is part of the diet. You'll also need to write a constraint to link them.)
 - b. Many people dislike celery and frozen broccoli. So at most one, but not both, can be selected.
 - c. To get day-to-day variety in protein, at least 3 kinds of meat/poultry/fish/eggs must be selected. [If something is ambiguous (e.g., should bean-and-bacon soup be considered meat?), just call it whatever you think is appropriate – I want you to learn how to write this type of constraint, but I don't really care whether we agree on how to classify foods!]

Methodology

- I. First, I created two variables: M and epsilon. We want to link the binary and continuous variables. I set M = 100 to make it large enough to not limit the max serving. I also set epsilon to 0.1 to enforce a reasonable minimum serving. I also prevented celery and broccoli from being selected at the same time. The constraint ≤ 1 ensures that either one is selected or none.
- II. Next, I scanned through the list of foods for keywords: beef, chicken, pork, turkey, fish, egg, tuna, ham, salmon, sardines, frankfurter, and bacon. Then, I filtered those into meat_foods and required that at least 3 different ones needed to be selected.
- III. Finally, I used prob.solve() to run the optimization.

Results

Optimal Diet Plan:

	Food	Servings	Cost per Serving	Total Cost
0	Celery, Raw	42.42	0.04	1.6968
1	Lettuce, Iceberg, Raw	82.67	0.02	1.6534
2	Oranges	3.09	0.15	0.4635
3	Poached Eggs	0.10	0.08	0.0080
4	Scrambled Eggs	0.10	0.11	0.0110
5	Bologna, Turkey	0.10	0.15	0.0150
6	Peanut Butter	1.96	0.07	0.1372
7	Popcorn, Air-Popped	13.21	0.04	0.5284

Total Cost: \$ 4.51

As a result, the optimal solution is when the value is \$4.51. This is the minimum total cost to meet all nutrient constraints. The cheapest combinations of foods that satisfies all daily nutrient minimum and maximum requirements is:

1. Celery, Raw: 42.42 servings
2. Lettuce, Iceberg, Raw: 82.67 servings
3. Oranges: 3.09 servings
4. Poached Eggs: 0.10 servings
5. Scrambled Eggs: 0.10 servings
6. Bologna, Turkey: 0.10 servings
7. Peanut Butter: 1.96 servings
8. Popcorn, Air-popped: 13.21 servings

Discussion of Results

In this result, celery and lettuce are still used heavily but controlled. This is better than the last model because the servings are linked to binary variables. The protein variety (poached eggs, scrambled eggs, bologna) meet the constraint 'at least 3'. The servings are small, but they meet the selection requirement. We can conclude that this model is more nutritionally valid and realistic while staying cheap.

References:

AI

https://mediaspace.gatech.edu/media/ISYE+6501+OMSA+Office+Hour/1_irykv0ab

<https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.iloc.html>

<https://coin-or.github.io/pulp/technical/pulp.html>